

Assignment no: 7

Aim: Develop a full stack web application using MERN stack to perform CRUD operations.

Implementation:

Step 1: Initialize npm project

```
PS C:\Users\rzamb\Desktop\Rishabh_Reference Books\T.Y. Sem 6\FSD-LAB\backend> npm init -y
Wrote to C:\Users\rzamb\Desktop\Rishabh_Reference Books\T.Y. Sem 6\FSD-LAB\backend\package.json:

{
  "name": "backend",
  "version": "1.0.0",
  "description": "",
  "main": "index.js",
  "scripts": {
    "test": "echo \"Error: no test specified\" && exit 1"
  },
  "keywords": [],
  "author": "",
  "license": "ISC",
  "type": "commonjs"
}
```

Step 2: Install Dependencies

```
PS C:\Users\rzamb\Desktop\Rishabh_Reference Books\T.Y. Sem 6\FSD-LAB\backend> npm install express mongoose cors body-parser

added 84 packages, and audited 85 packages in 8s

24 packages are looking for funding
  run `npm fund` for details
```

Step 3: Create main file ([server.js](#))

Name	Date modified	Type	Size
node_modules	09-04-2026 13:46	File folder	
package	09-04-2026 13:46	JSON Source File	1 KB
package-lock	09-04-2026 13:46	JSON Source File	37 KB
server	09-04-2026 13:48	JavaScript Source ...	2 KB

Step 4: Update main file

```
JS server.js U X
C: > Users > rzamb > Desktop > Rishabh_Reference Books > T.Y. Sem 6 > FSD-LAB > backend > JS server.js > ...
1  const express = require("express");
2  const mongoose = require("mongoose");
3  const cors = require("cors");
4
5  const app = express();
6  app.use(express.json());
7  app.use(cors());
8
9  // MongoDB Connection
10 mongoose.connect("mongodb://127.0.0.1:27017/studentDB", {
11   useNewUrlParser: true,
12   useUnifiedTopology: true
13 });
14
15 // Schema
16 const studentSchema = new mongoose.Schema({
17   firstName: String,
18   lastName: String,
19   rollNo: String,
20   password: String,
21   contact: String
22 });
23
24 const Student = mongoose.model("Student", studentSchema);
```

Step 5: Start MongoDB

```
PS C:\Users\rzamb> mongod
{"t":{"$date":"2026-04-09T13:50:23.071+05:30"},"s":"I", "c":"CONTROL", "id":23285, "ctx":"thread1","msg":"Automatically disabling TLS 1.0, to force-enable TLS 1.0 specify --sslDisabledProtocols 'none'"}
{"t":{"$date":"2026-04-09T13:50:23.077+05:30"},"s":"I", "c":"CONTROL", "id":5945603, "ctx":"thread1","msg":"Multi threading initialized"}
```

Step 6: Run Main Server File

```
Windows PowerShell
PS C:\Users\rzamb\Desktop\Rishabh_Reference Books\T.Y. Sem 6\FSD-LAB\backend> node server.js
Server running on port 5000
```

Step 7: Update Frontend [App.js](#) and run it

```
JS App.js U X
C: > Users > rzamb > Desktop > Rishabh_Reference Books > T.Y. Sem 6 > FSD-LAB > frontend > src > JS App.js > default
1 import React, { useState, useEffect } from "react";
2
3 function App() {
4   const [student, setStudent] = useState({
5     firstName: "",
6     lastName: "",
7     rollNo: "",
8     password: "",
9     contact: ""
10  });
11
12   const [data, setData] = useState([]);
13
14   const handleChange = (e) => {
```

Step 8: Test application

```
localhost:3000
```

Student Management System

Hiralal	Chandan	99	1213	782105
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Add Student

Student List

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- Hiralal Chandan (99) - 782105